OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

14.7 Signal boards

Signals are not always separate signs on a single sheet of metal. Several signs can be coupled on one board. They can be of the following types:

- static board of @type="staticBoard"
- variable message board of @type="vmsBoard"
- multi board with static and dynamic parts of @type="multiBoard"

14.7.1 Static boards

Elements in UML model

`<staticBoard>` element

```
UML class:  t_road_signals_staticBoard
XML tag:    <staticBoard> (Multiplicity: 0..*)
Introduced: 1.8.0
```

A `<signal>` element that contains a `<staticBoard>` element. The signs that are displayed on a static board are defined as separate `<sign>` elements.

`<sign>` element

```
UML class:  t_road_signals_board_sign
XML tag:    <sign> (Multiplicity: 0..*)
```

```
Introduced: 1.8.0
```

A `<sign>` element on a static board defined in the local coordinate system of the `<signal>` element. A `<sign>` element may have all attributes and child elements of a signal.

Table 128. Attributes of the <sign> element

Name	Type	Use	Unit	Introduced	Description
`countryRevision`	string	optional			Defines the year of the applied traffic rules
`country`	e_countryCode	optional			Country code of the road, see ISO 3166-1, alpha-2 codes.
`dynamic`	t_yesNo	required			Indicates whether the signal is dynamic or static. Example: traffic light is dynamic
`hOffset`	double	optional	rad		Heading offset of the signal (relative to @orientation, if orientation is equal to “+” or “-“) Heading offset of the signal (relative to road reference line, if orientation is equal to “none”)
`height`	t_grEqZero	optional	m		Height of the signal, measured from bottom edge of the signal.
`id`	string	required			Unique ID of the signal within the OpenDRIVE file
`length`	t_grEqZero	optional	m	1.8.0	Length of the signal’s bounding box. @length is defined in the local coordinate system u/v along the u-axis
`name`	string	optional			Name of the signal. May be chosen freely.
`orientation`	e_orientation	required			"+" = valid in positive s- direction "-" = valid in negative s- direction "none" = valid in both directions
`pitch`	double	optional	rad		Pitch angle of the signal, relative to the inertial system (xy-plane)

Name	Type	Use	Unit	Introduced	Description
`roll`	double	optional	rad		Roll angle of the signal after applying pitch, relative to the inertial system (x”y”-plane)
`subtype`	string	required			Subtype identifier according to country code or "-1" / "none"
`text`	string	optional			Additional text associated with the signal, for example, text on city limit "City\nBadAibling"
`type`	string	required			Type identifier according to country code or "-1" / "none". See extra document.
`unit`	e_unit	optional			Unit of @value
`v`	double	required	m	1.8.0	Local v-coordinate of the sign on the board
`value`	double	optional			Value of the signal, if value is given, unit is mandatory
`width`	t_grEqZero	optional	m		Width of the signal’s bounding box. @width is defined in the local coordinate system u/v along the v-axis
`z`	double	required	m	1.8.0	Local z-coordinate of the sign on the board

XML example

```
<signal s="4.0"
        t="1.0"
        id="534"
        name="board"
        dynamic="no"
        orientation="+"
        zOffset="5.00"
        country="OpenDRIVE"
        type="staticBoard"
        subtype="-1"
        hOffset="0"
        pitch="0"
        roll="0"
        height="2.0"
        width="1.5">
    <validity from="-2" to="-2"/>
    <staticBoard>
        <sign id="535" Country="DE" type="274" subtype="60" countryRevision="2017" v="-0.5" z="1.5" width="0.5" hei
            <validity from="-2" to="-2"/>
            <signalDependency id ="536"/>
            <signalDependency id ="537"/>
        </sign>
        <sign id="536" Country="DE" type="1010" subtype="51" countryRevision="2017" v="-0.75" z="0.9" width="0.420"
        <sign id="537" Country="DE" type="1040" subtype="30" countryRevision="2017" v="-0.75" z="0.6" width="0.420"
        <sign id="538" Country="DE" type="1012" subtype="36" countryRevision="2017" v="-0.75" z="0.3" width="0.420"
        <sign id="539" Country="DE" type="274" subtype="80" countryRevision="2017" v="0.75" z="1.5" width="0.420" h
            <signalDependency id ="540" />
        </sign>
        <sign id="540" Country="DE" type="1040" subtype="30" countryRevision="2017" v="-0.75" z="0.6" width="0.420"
        <sign id="541" Country="DE" type="1012" subtype="36" countryRevision="2017" v="-0.75" z="0.3" width="0.420"
    </staticBoard>
 </signal>
```

```
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0
3
1
3
2
```

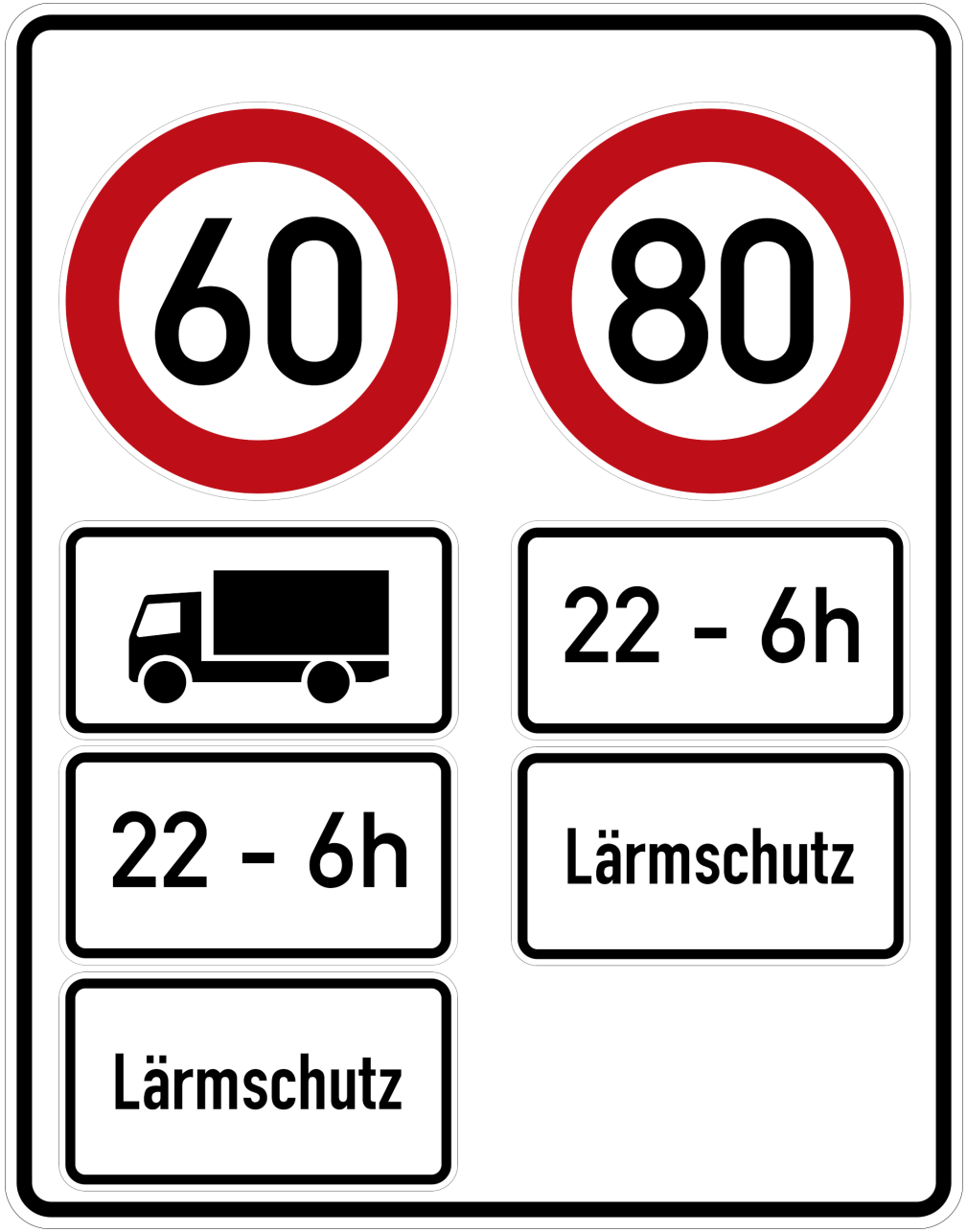

Figure 134. multiStaticSign from XML example above

Rules

- asam.net:xodr:1.8.0:road.signal.boards.static_board_use_correct_type: Static signal boards shall be specified to be @type="staticBoard".
- Static signal boards shall be specified to be @dynamic="false".
- The `<validity>` element of a `<sign>` element shall override the `<validity>` element of the parent `<signal>` element.
- The `<signalDependency>` element of a `<sign>` element shall override the `<signalDependency>` element of the parent `<signal>` element.
- Static boards shall not be used for single signals, for example, a stop sign on a single sheet of metal.

Related topics

- Section 14.3, "Signal dependency"
- Section 14.4, "Signal reference"
- Section 14.8, "Signal semantics"
- Section 14.7.2, “Variable message boards (VMS)”
- Section 14.7.3, “Multi boards”
- Section 14.7.4, “Gantry”

14.7.2 Variable message boards (VMS)

Elements in UML model

`<vmsBoard>` element

```
UML class:  t_road_signals_vmsBoard
XML tag:    <vmsBoard> (Multiplicity: 0..*)
Introduced: 1.8.0
```

Variable message boards can change their values during the simulation in ASAM OpenSCENARIO.

Variable message boards are switched off if they are not specified in ASAM OpenSCENARIO.

Table 129. Attributes of the <vmsBoard> element

Name	Type	Use	Unit	Introduced	Description
`displayHeight`	double	optional	m	1.8.0	Height of the display
`displayType`	e_road_signals_displayType	required		1.8.0	Functional type of the display
`displayWidth`	double	optional	m	1.8.0	Width of the display
`v`	double	required	m	1.8.0	Local v-coordinate of the board
`z`	double	required	m	1.8.0	Local z-coordinate of the board

`<displayArea>` element

```
UML class:  t_road_signals_displayArea
XML tag:    <displayArea> (Multiplicity: 0..*)
Introduced: 1.8.0
```

A display area is the recommended position of the signal to be visualized in the simulation. A display area is specified in the `<displayArea>` element. A `<displayArea>` element is defined in the local coordinate system of the `<signal>` element. The @index attribute can be used in ASAM OpenSCENARIO to reference the display area. In ASAM OpenSCENARIO a different local display area position may be specified.

Table 130. Attributes of the <displayArea> element

Name	Type	Use	Unit	Introduced	Description
`height`	t_grEqZero	required	m	1.8.0	Height of the `<displayArea>`

Name	Type	Use	Unit	Introduced	Description
`index`	int	required		1.8.0	Index of the `<displayArea>`
`v`	double	required	m	1.8.0	Local v-coordinate of the `<displayArea>` on the board
`width`	t_grEqZero	required	m	1.8.0	Width of the `<displayArea>`
`z`	double	required	m	1.8.0	Local z-coordinate of the `<displayArea>` on the board

XML example

```xml
<signal s="4.0"
        t="1.0"
        id="534"
        name="board"
        dynamic="yes"
        orientation="+"
        zOffset="5.00"
        country="OpenDRIVE"
        type="vmsBoard"
        subtype="-1"
        hOffset="0"
        pitch="0"
        roll="0"
        height="1.5"
        width="1.5">
    <vmsBoard displayHeight="1.5" displayWidth="1.5" material="colorGraphics">
        <displayArea index="1" v="7" z="3" width="1.4" height="1.4">
            <validity from="-2" to="-2" />
        </displayArea>
        <displayArea index="2" v="5.5" z="3" width="1.4" height="1.4">
            <validity from="-2" to="-3" />
        </displayArea>
        <displayArea index="3" v="5.5" z="0.5" width="1.4" height="0.4" >
            <validity from="-3" to="-3" />
        </displayArea>
    </vmsBoard>
</signal>
```

```
3
2
4
2
5
2
6
2
7
```

Rules

- Variable message boards shall be specified to be @type="vmsBoard".
- Variable message boards shall be specified to be @dynamic="true".
- The `<validity>` element of a `<displayArea>` element shall override the `<validity>` element of the parent `<signal>` element.
- The `<signalDependency>` of a `<displayArea>` element shall override the `<signalDependency>` element of the parent `<signal>` element.

Related topics

- Section 14.7.1, “Static boards”
- Section 14.7.3, “Multi boards”
- Section 14.7.4, “Gantry”

14.7.3 Multi boards

A multi board is a board that consists of static boards and variable message boards. The size of the parent `<signal>` element covers all static and variable message boards.

XML example

```
<signal s="4.0"
        t="1.0"
        id="534"
        name="board"
        dynamic="yes"
        orientation="+"
        zOffset="5.00"
        country="OpenDRIVE"
        type="multiBoard"
        subtype="-1"
        hOffset="0"
        pitch="0"
        roll="0"
        height="3.0"
        width="7.5">
    <vmsBoard displayHeight="2.5" displayWidth="7.0" material="colorGraphics" v="0" z="0.5">
        <displayArea index="1" v="7" z="3" width="1.4" height="1.4">
            <validity from="-2" to="-2" />
        </displayArea>
        <displayArea index="2" v="5.5" z="3" width="1.5" height="1.5">
            <validity from="-2" to="-3" />
        </displayArea>
        <displayArea index="3" v="5.5" z="0.5" width="1.5" height="0.5">
            <validity from="-3" to="-3" />
        </displayArea>
    </vmsBoard>
    <staticBoard>
        <sign id="535" country="DE" type="386" subtype="32" countryRevision="2017" v="-7" z="0.2" width="0.5" heigh
            <validity from="-2" to="-2" />
        </sign>
        <sign id="535" country="DE" type="405" subtype="-1" countryRevision="2017" v="-5" z="0.2" width="0.5" heigh
            <validity from="-2" to="-3" />
        </sign>
        <sign id="535" country="DE" type="386" subtype="32" countryRevision="2017" v="-3.5" z="0.2" width="0.5" hei
            <validity from="-3" to="-3" />
        </sign>
```

```
    </staticBoard>
</signal>
```

Rules

- asam.net:xodr:1.8.0:road.signal.boards.multi_board_have_sub_boards: A multi board shall have at least one static signal board and at least one variable message board.

- asam.net:xodr:1.8.0:road.signal.boards.multi_board_use_correct_type: Multi boards shall be specified to be @type="multiBoard".
- asam.net:xodr:1.8.0:road.signal.boards.multi_board_use_dynamic_true: Multi boards shall be specified to be @dynamic="true".

Related topics

- Section 14.7.1, “Static boards”
- Section 14.7.2, “Variable message boards (VMS)”
- Section 14.7.4, “Gantry”

14.7.4 Gantry

Elements in UML model

`<vmsGroup>` element

```
UML class:  t_signalGroup_vmsGroup
XML tag:    <vmsGroup> (Multiplicity: 0..*)
Introduced: 1.8.0
```

On a gantry there can be one large variable message board or several smaller variable message boards. ASAM OpenSCENARIO requires to treat a gantry that has one large variable message board or several smaller variable message boards the same way. Therefore variable message boards that are on the same gantry shall be grouped and their indexes shall be redefined if not unique.

Table 131. Attributes of the <vmsGroup> element

Name	Type	Use	Introduced	Description
`id`	string	required	1.8.0	Unique id of the `<vmsGroup>`

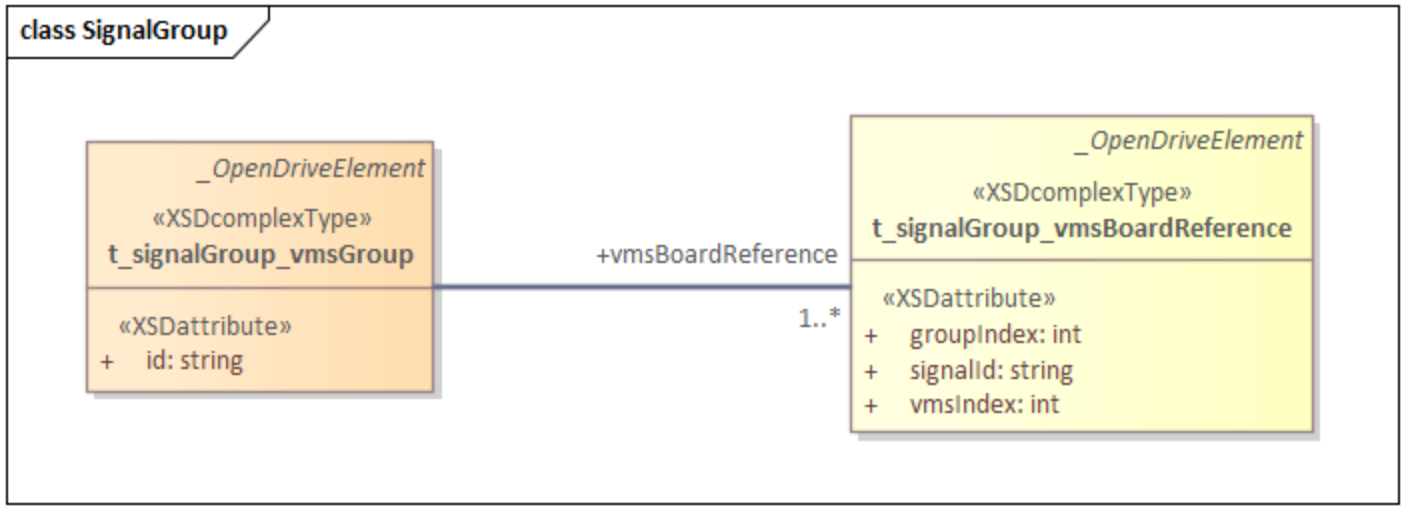

Figure 135. UML class diagram of the SignalGroup class

Figure 135 shows the UML class diagram of the ASAM OpenDRIVE SignalGroup class.

`<vmsBoardReference>` element

```
UML class:  t_signalGroup_vmsBoardReference
XML tag:    <vmsBoardReference> (Multiplicity: 1..*)
Introduced: 1.8.0
```

Variable message board references list all variable message boards that belong to the same gantry.

Table 132. Attributes of the <vmsBoardReference> element

Name	Type	Use	Introduced	Description
`groupIndex`	int	required	1.8.0	groupIndex the index of the grouped boards shall be unique within the `<vmsGroup>`
`signalId`	string	required	1.8.0	Id of the signal that has a `<vmsBoard>` assigned
`vmsIndex`	int	required	1.8.0	vmsIndex, the index of the `<vmsBoard>`

XML example

```
<vmsGroup id="27" >
    <vmsBoardReference signalId="501" vmsIndex="1" groupIndex="1" />
    <vmsBoardReference signalId="502" vmsIndex="1" groupIndex="2" />
    <vmsBoardReference signalId="503" vmsIndex="1" groupIndex="3" />
</vmsGroup>
```

Rules

- Each gantry shall have one `<vmsGroup>` element with at least one `<vmsBoardReference>` element.
- All variable message boards within a `<vmsGroup>` element shall belong to the same gantry.

Related topics

- Section 14.7.1, “Static boards”
- Section 14.7.2, “Variable message boards (VMS)”
- Section 14.7.3, “Multi boards”